

Discrete Mathematics

CS1112

Lecture 6

Logic

Propositions

- A proposition is “A statement that can either be true or false but not both”
- The following are propositions
 - $5 + 3 = 7$
 - The desert is green
 - The sky is blue
- The following are not propositions
 - How old are?
 - Do not drive too fast

Propositions

- A proposition is usually denoted using letters p , q , r , ... -- a variable that can take one of two values
 - True $\rightarrow T$
 - False $\rightarrow F$

Propositions

- Two types of propositions
 - simple
 - Atomic statements that are indivisible
 - Compound
 - Formed by combining simple or compound propositions using connectives
- Connectives
 - Negation $\neg$, conjunction $\wedge$, disjunction $\vee$, exclusive or $\oplus$, implication $\rightarrow$,

Negation

- Let p be a proposition, the negation of p denoted $\neg p$ (or $\sim p$) is a proposition representing the statement “It is not the case that p ”.
- Example: If $p = \text{“Ahmed is now here”}$ then $\neg p = \text{“Ahmed is not here”}$
- If $p = T$ the $\neg p = F$
- If $p = F$ the $\neg p = T$

Negation

- Truth table for $\neg p$

p	$\neg p$
T	F
F	T

Conjunction

- Let p and q be two propositions, the conjunction of p and q is a proposition “ p and q ”, denoted as $p \wedge q$
- $p \wedge q$ is true when both p and q are true and false otherwise
- If p is true and q is true then $p \wedge q$ will be true
 - In all other cases $p \wedge q$ will be false

Conjunction

- Truth table for $p \wedge q$

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction

- Let p and q be two propositions, the disjunction of p and q , denoted $p \vee q$, is the proposition “ p or q ”
- $p \vee q$ is true if any of p or q is true, if both are true the $p \vee q$ is also true.
 - $p \vee q$ is false only if both p and q are false.
- If both p and q are true the $p \vee q$ is also true
 - $\rightarrow$ it is sometimes called “inclusive or”

Disjunction

- Truth table for $p \vee q$

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Exclusive or ($\oplus$)

- Let p and q be two propositions, the exclusive or of p and q , denoted $p \oplus q$ is a proposition that is true only if either p or q is true but not both.
- If both p and q are true the $p \oplus q$ is false
- If both p and q are false the $p \oplus q$ is false

Exclusive or $\oplus$

- Truth table for $p \oplus q$

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Conditional Statements

- Let p and q be two propositions, the conditional statement or implication $p \rightarrow q$ is the proposition “if p then q ”,
 - $P \rightarrow q$ is read “ p implies q ”
- $p \rightarrow q$ is false when p is true and q is false and is true otherwise.
- p is called the hypothesis, antecedent or premise
- q is called the conclusion or the consequence

Conitional Statements

- Truth table for $p \rightarrow q$

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Conditional Statements

- Example

- Let $p = \text{“it is raining”}$ and $q = \text{“it is cloudy”}$,

- $p \rightarrow q$ is the statement “if it is raining then it is cloudy”
 - $q \rightarrow p$ is the statement “if it is cloudy then it is raining”

Conditional Statements

- There are many ways to express $p \rightarrow q$ such as
 - If p then q
 - q if p
 - If p , q
 - p is sufficient for q
 - q when p
 - p implies q
 - ...

BiConditional Statements

- Let p and q be two propositions, the bi-conditional statement $p \leftrightarrow q$ is the statement “ p if and only if q ”, that is p and q are logically equivalent.
- $p \leftrightarrow q$ is true when both p and q are true or both are false and false otherwise.

BiConditional Statements

- Truth table for $p \leftrightarrow q$

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

BiConditional Statements

- $p \leftrightarrow q$ has the following forms
 - “p iff q”
 - “p is necessary and sufficient for q”
 - “p is equivalent to q”
 -
- Example
 - $x > y$ if and only if $y < x$
 - “ $x > y$ ” $\leftrightarrow$ “ $y < x$ ”

Inverse, Converse & Contrapositive

- Let p and q be two propositions, we can construct three propositions from the conditional proposition $p \rightarrow q$
 - The converse is $q \rightarrow p$
 - The inverse is $\neg p \rightarrow \neg q$
 - The contrapositive is $\neg q \rightarrow \neg p$
 - The contrapositive has the same truth values as the $p \rightarrow q$

Inverse, Converse & Contrapositive

- Example
 - “if I am in Riyadh then I am in KSA” $p \rightarrow q$
 - Inverse: “If I am not in Riyadh then I am not in KSA”
 - Converse: “if I am in KSA then I am in Riyadh”
 - Contrapositive: “if I am not in KSA then I am not in Riyadh”
- Clearly the contrapositive is equivalent to the original $p \rightarrow q$

Inverse, Converse & Contrapositive

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$q \rightarrow p$	$\neg p \rightarrow \neg q$	$\neg q \rightarrow \neg p$
T	T	F	F	T	T	T	T
T	F	F	T	F	T	T	F
F	T	T	F	T	F	F	T
F	F	T	T	T	T	T	T

Laws of Logic

Law		Name
$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$ $p \rightarrow q \equiv \neg p \vee q$ $\neg \neg p \equiv p$		Equivalence Implication Double negation
$p \wedge p \equiv p$	$p \vee p \equiv p$	Idempotent
$p \wedge q \equiv q \wedge p$	$p \vee q \equiv q \vee p$	Commutative
$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	$(p \vee q) \vee r \equiv p \vee (q \vee r)$	Associative
$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$	Distributive
$\neg(p \wedge q) \equiv \neg p \vee \neg q$	$\neg(p \vee q) \equiv \neg p \wedge \neg q$	DeMorgan's
$p \wedge T \equiv p$	$p \vee F \equiv p$	Identity
$p \wedge F \equiv F$	$p \vee T \equiv T$	Annihilation
$p \wedge \neg p \equiv F$	$p \vee \neg p \equiv T$	Inverse
$p \wedge (p \vee q) \equiv p$	$p \vee (p \wedge q) \equiv p$	Absorption

Tautology and Contradiction

- A compound proposition that is always true no matter what are the truth values of the propositional variables that occur in it, is called a tautology.
- A compound proposition that is always false is called a contradiction.

Tautology

p	q	$p \rightarrow q$	$(p \rightarrow q) \wedge p$	$((p \rightarrow q) \wedge p) \rightarrow q$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

Contradiction

p	$\neg p$	$p \wedge \neg p$
T	F	F
F	T	F

Logical Equivalence

- Two compound propositions are said to be logically equivalent if they have the same truth values whatever the values of their combining propositions>

Example:

Let A = “Today is Monday and the Weather is fine”

B = “The weather is fine and either today is Monday or the weather is not fine”

Logical Equivalence

Let $A = \text{“Today is Monday and the Weather is fine”}$

$B = \text{“The weather is fine and either today is Monday or the weather is not fine”}$

put $p = \text{“Today is Monday”}$ and $q = \text{“The weather is fine”}$

Then $A = p \wedge q$ and $B = q \wedge (p \vee \neg q)$

The following truth table show the equivalence of A and B:

Logical Equivalence

p	q	$\neg q$	$p \vee \neg q$	$p \wedge q$	$q \wedge (p \vee \neg q)$
T	T	F	T	T	T
T	F	T	T	F	F
F	T	F	F	F	F
F	F	T	T	F	F

Logical Equivalence

Logical equivalence can be proved also using laws of logic.

Goal is to prove that $p \wedge q = q \wedge (p \vee \neg q)$

LHS: $q \wedge (p \vee \neg q)$

$(q \wedge p) \vee (q \wedge \neg q)$

Distributive law

$(q \wedge p) \vee (F)$

Inverse law

$(q \wedge p)$

Identity law

$(p \wedge q)$

commutative law

$=$ RHS. Hence $p \wedge q = q \wedge (p \vee \neg q)$

Predicates

- The statement: “ $x > 5$ ” is an example of a predicate.
- A predicate is a statement which contain one or more variables; it cannot be Assigned a truth value until the value of the variables are specified.
- e.g. if $x=7$, the sentence is true, but if $x=2$, it is false.

Quantifiers

- Let $P(x)$ be the statement “ x submitted the homework” and $A =$ set of students in this section, then:
- $\forall x \in A, P(x)$ is the statement “all students in this section submitted the homework”

$\forall$ is read “for all” and called the universal quantifier

Quantifiers

- $\exists$ is read “there exists” and called the existential quantifier.
- $\exists x \in A, P(x)$ is the statement “there is a students in this section submitted the homework”

for $\exists x \in A, P(x)$ to be true it must be true for at least one element of the set A .

for $\forall x \in A, P(x)$ to be true it must be true for ALL elements of the set A .

Quantifiers

Examples:

$$\forall x \in \mathbb{N}, x > -5$$

$$\exists x \in \mathbb{N}, x \text{ divides } 7$$

$$\exists d \in \text{WeekDays}, d \text{ is a public holiday.}$$

$$\forall x \in \mathbb{R}, (x + 5)/3 \in \mathbb{R}$$

$$\forall x \in \{1, 2, 3, \dots, 10\}, x \text{ do not divide } 11$$

$$\exists x \in \{1, 2, 3, \dots, 10\}, x \text{ divides } 24$$

$$\exists x \in \{1, 2, 3, \dots, 10\}, x \text{ divides } 49$$